

FILTER DESIGN TEST CASE

Lattice Filter - Band 26 (850 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Sarah Okonkwo
Reviewer:	Ana Reyes
Design Center:	Helsinki 5G Lab
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Priority:	P4 - Low

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting Band 26 (850 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on AIN on Si substrate with a target center frequency of 4311.5 MHz and bandwidth of 325.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: Band 26 (850 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: AIN on Si
- Package: LGA (Land Grid Array)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the Lattice Filter design targeting Band 26 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4311.5 MHz
Bandwidth (3 dB):	325.7 MHz
Insertion Loss Target:	<= 3.6 dB
Return Loss Target:	>= 13.0 dB
Out-of-Band Rejection:	>= 22.0 dB
Isolation (Duplexer):	>= 36.0 dB
Q Factor:	5618
Temperature Coefficient:	-22.8 ppm/°C
Die Size:	0.6 x 1.4 mm
Wafer Size:	6-inch
Number of Resonators:	6
Electrode Material:	Mo/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Lattice Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 4311.5 MHz, BW 325.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.6 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 12934 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.6 dB
- Return loss in passband shall be >= 13.0 dB
- Out-of-band rejection at +/- 489 MHz offset >= 22.0 dB
- Group delay variation across passband shall be <= 9 ns
- Temperature drift over -40°C to +85°C shall be <= 12.3 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2025-03-16
Devices Tested: 69
Insertion Loss (meas): 3.26 dB
Return Loss (meas): 14.9 dB
Rejection (meas): 25.2 dB
Isolation (meas): 40.7 dB
Q Factor (meas): 5618
Group Delay Variation: 7.9 ns
Power Handling: +26 dBm
Die Yield: 82.5%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated below-target performance for Band 26 (850 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

Author:	Sarah Okonkwo	Date:	2025-03-16
Reviewer:	Ana Reyes	Date:	2025-03-17
Approver:	Dr. Hans Weber	Date:	2025-03-26

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